PLSC 40601

Week 3: Trees and forests.

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Spring 2024

Housekeeping

- ?

Trees

Regression Trees

- Suppose we have joint data, (Y, X_1, X_2).

Regression Trees

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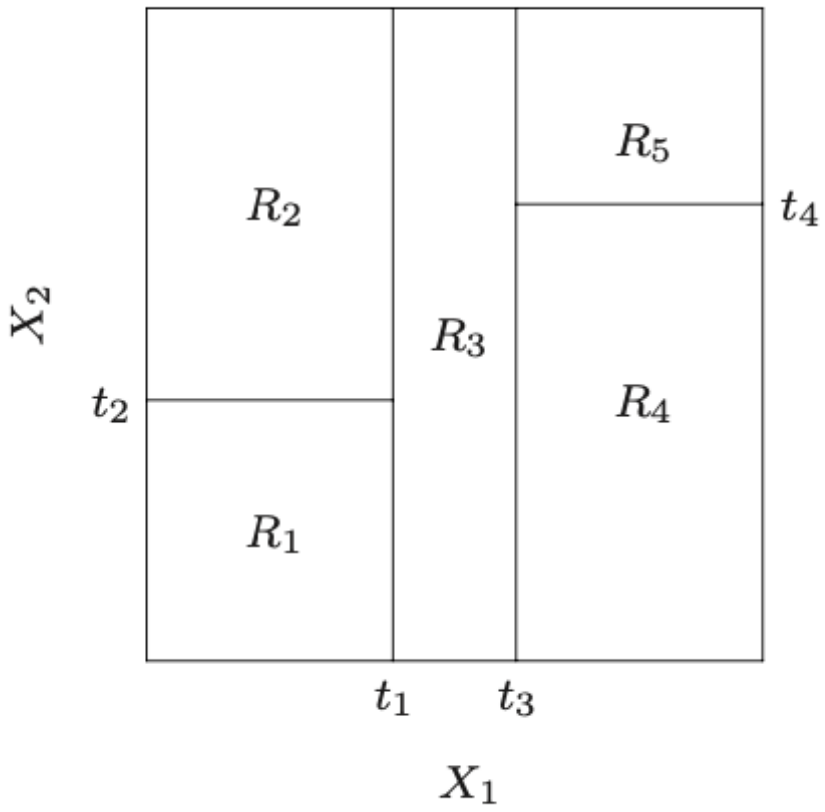
- Our goal is to partition the data with the objective of prediction.



Hastie et al. (2009)

Regression Trees

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$$\hat{c}_m = \bar{y}_{\mathrm{x}_i \in R_m}$$

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$$\min_{j,s} \left[\min_{c_1} \sum_{i:x_{j[i]} \in R_1(j,s)} (y_i - c_1)^2 + \min_{c_2} \sum_{i:x_{j[i]} \in R_2(j,s)} (y_i - c_2)^2 \right]$$

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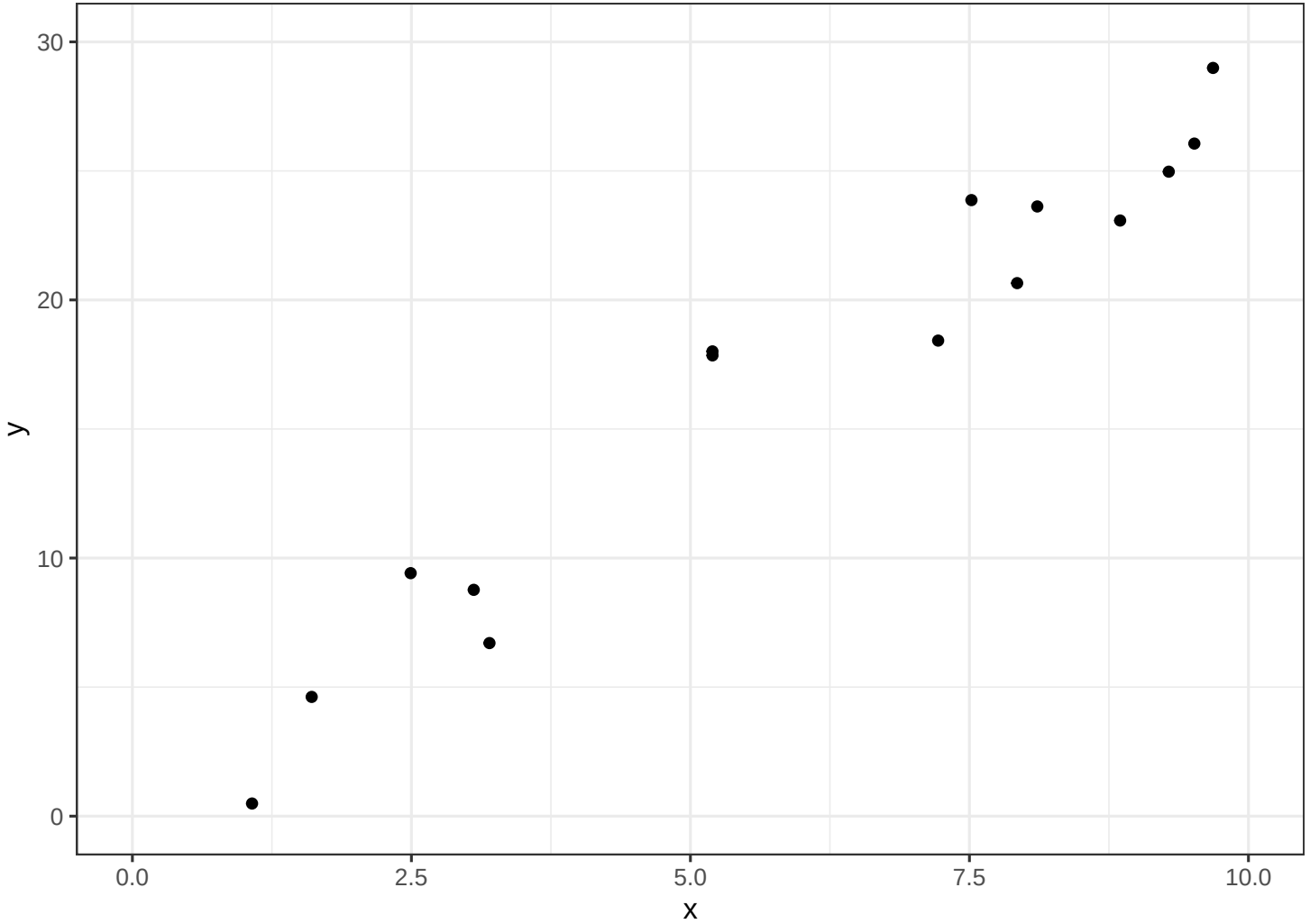
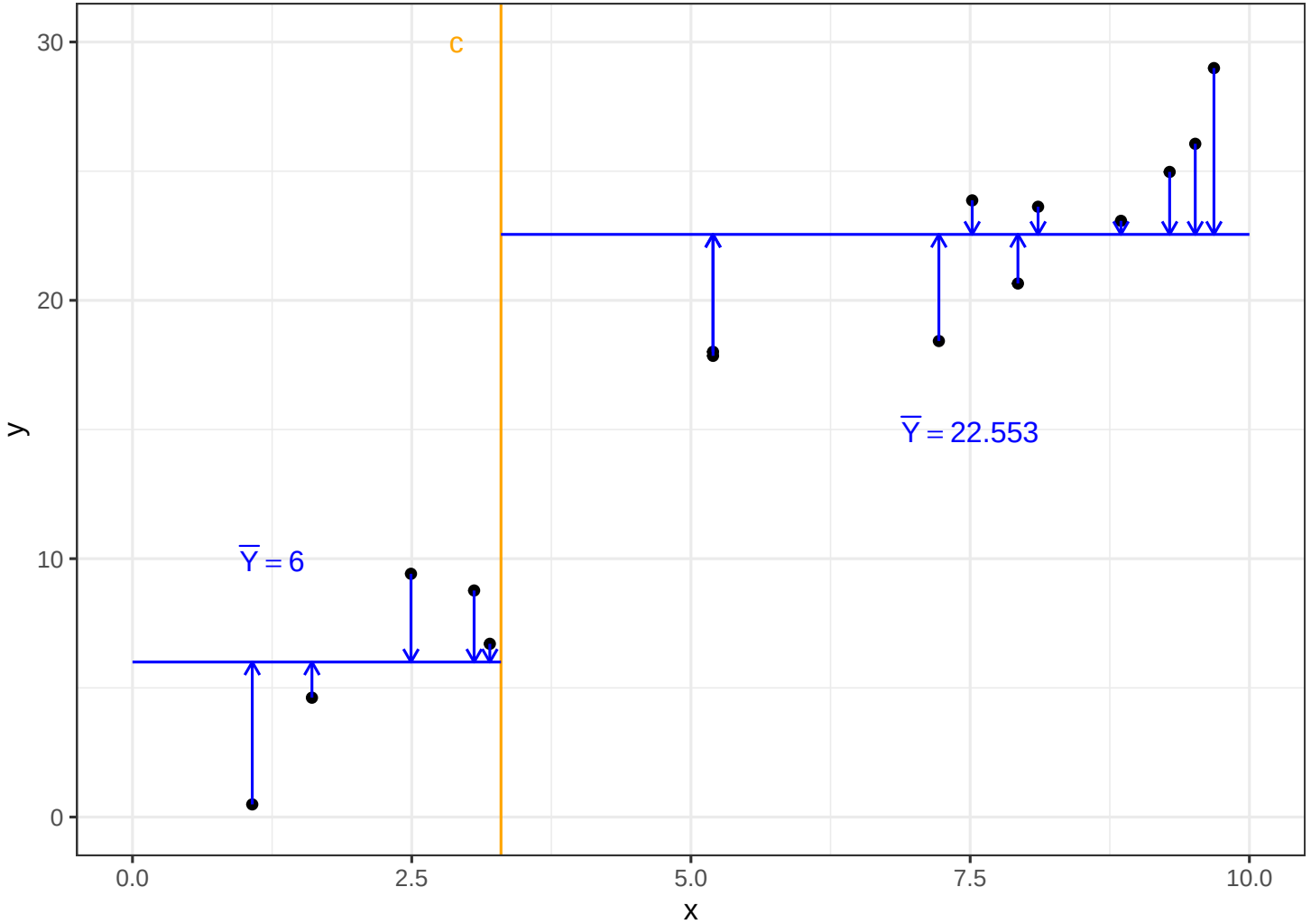
  - Then pick s to solve the outer minimization problem for a given variable j.

Regression Trees

- For just one variable:


30
20
10
0
y
0.0
2.5
5.0
7.5
10.0
x


c
$\overline{Y} = 6$
$\overline{Y} = 22.553$
y
x
30
20
10
0
0.0
2.5
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10.0

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- We will do the same approach to finding thresholds to minimize prediction error, but we'll want to pick which X_k we use for threshholding, as well.
- Generally, we'll define the depth of the tree as 2 or three variables; first we'll split on X_k, then we'll split on X_j...

Elements of trees

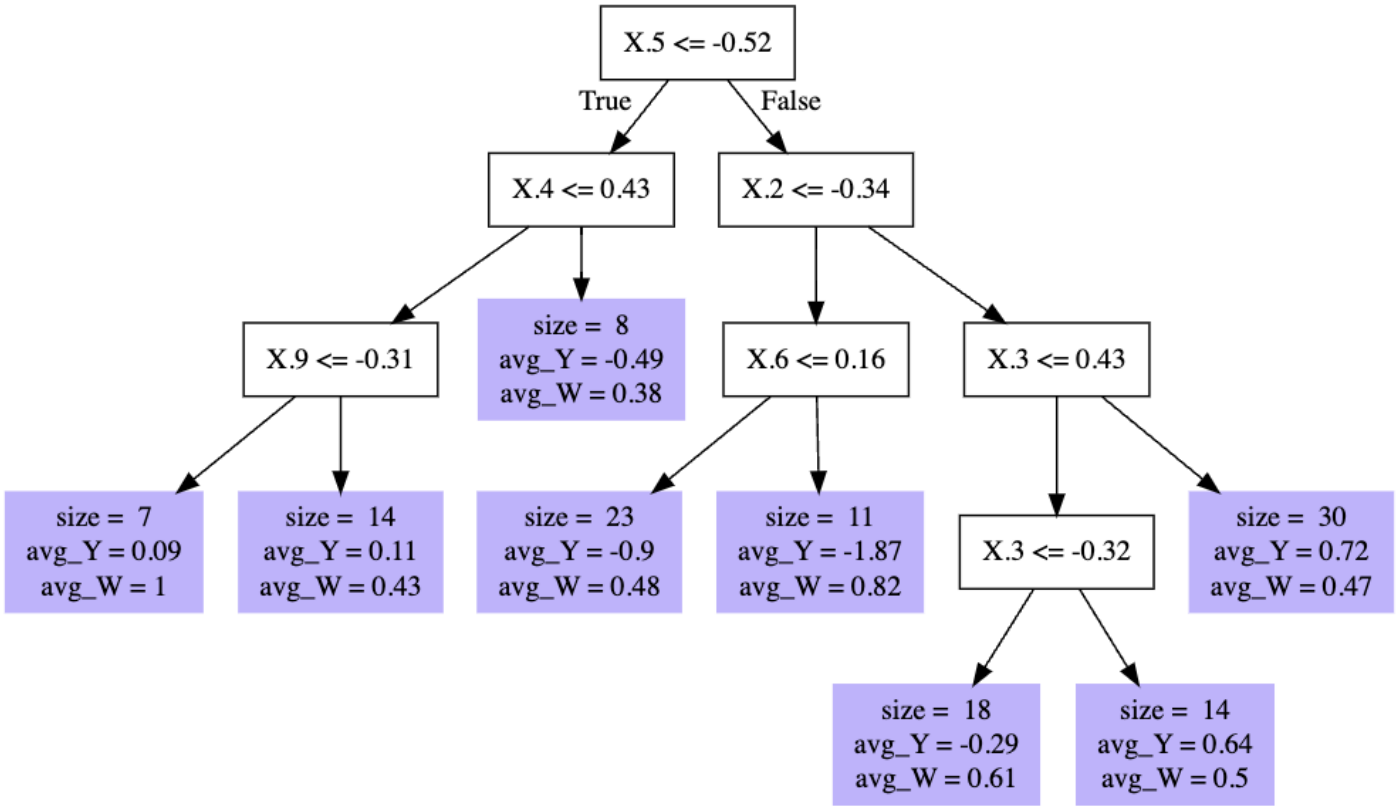
```
> n <- 500
> p <- 10
> X <- matrix(rnorm(n * p), n, p)
> W <- rbinom(n, 1, 0.5)
> Y <- pmax(X[, 1], 0) * W + X[, 2] +
+   pmin(X[, 3], 0) + rnorm(n)
> c.forest <- causal_forest(X, Y, W)
> tree <- get_tree(c.forest, 1)
> leaf.nodes <- get_leaf_node(tree, X[1:5, ])
```

Elements of trees

```
> tree

GRF tree object
Number of training samples: 250
Variable splits:
(1) split_variable: X.5  split_value: -0.520294
  (2) split_variable: X.4  split_value: 0.426613
    (4) split_variable: X.9  split_value: -0.314623
      (8) * num_samples: 7  avg_Y: 0.09 avg_W: 1
      (9) * num_samples: 14  avg_Y: 0.11 avg_W: 0.43
    (5) * num_samples: 8  avg_Y: -0.49 avg_W: 0.38
  (3) split_variable: X.2  split_value: -0.344787
    (6) split_variable: X.6  split_value: 0.163222
      (10) * num_samples: 23  avg_Y: -0.9 avg_W: 0.48
      (11) * num_samples: 11  avg_Y: -1.87 avg_W: 0.82
    (7) split_variable: X.3  split_value: 0.428785
      (12) split_variable: X.3  split_value: -0.322743
        (14) * num_samples: 18  avg_Y: -0.29 avg_W: 0.61
        (15) * num_samples: 14  avg_Y: 0.64 avg_W: 0.5
      (13) * num_samples: 30  avg_Y: 0.72 avg_W: 0.47
```

A tree


X.5 <= -0.52
True
False
X.4 <= 0.43
X.2 <= -0.34
X.9 <= -0.31
size = 8
avg_Y = -0.49
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Elements of trees

- Node

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- Split/branches

Elements of trees

- Node
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- Leaves

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- What about when Y is a category?

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- For binary classification, minimize surrogate for classification error with split point s, $I(s) = \sum_{t=1}^{2} \gamma_t$

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- We can extend these methods to more complex classification tasks.

Policy trees, or decision trees

- Policy trees solve:

$$\pi^* = \underset{\pi \in \Pi}{\operatorname{argmax}} \left[\frac{1}{N} \sum_{i=1}^{N} \Gamma_i \left(\pi(X_i) \right) \right]$$

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$$\Gamma_i = \frac{1}{N} \sum_{i=1}^{N} \underbrace{\hat{\mathrm{E}}\left[Y_i | W_i = w, X_i \right]}_{\text{estimated outcome model}} + \underbrace{\frac{1}{N} \sum_{i=1}^{N} \frac{\mathbb{1}\{W_i = w\} \left(Y_i - \hat{\mathrm{E}}\left[Y_i | W_i = w, X_i \right] \right)}{e_i(w; s)}}_{\text{residual bias correction}}$$

[Recall]

- Estimating average treatment effects.

$$\mu(w, x) = \mathrm{E}\left[Y_i | W_i = w, \mathrm{X}_i = x\right]$$
$$e(x) = \mathrm{E}\left[W_i | \mathrm{X}_i = x\right]$$

$$\begin{aligned}
\tau &= \mathrm{E}\left[\mu(1, \mathrm{X}_i) - \mu(0, \mathrm{X}_i)\right] \\
&= \mathrm{E}\left[\frac{Y_i W_i}{e(\mathrm{X}_i)} - \frac{Y_i(1 - W_i)}{1 - e(\mathrm{X}_i)}\right] \\
&= \mathrm{E}\left[\frac{\left(Y_i - \mu(1, \mathrm{X}_i)\right) W_i}{e(\mathrm{X}_i)} - \frac{\left(Y_i - \mu(0, \mathrm{X}_i)\right)(1 - W_i)}{1 - e(\mathrm{X}_i)}\right] \\
&\qquad + \mathrm{E}\left[\mu(1, \mathrm{X}_i) - \mu(0, \mathrm{X}_i)\right]
\end{aligned}$$

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- Pruning?

Forests

Random forests…

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- Θ_k are i.i.d. random vectors (putting the *random* in random forests)
- With predictors or classifiers $\{h(\mathrm{x}, \Theta_k),\ k = 1, \ldots, K\}$, combine across trees, where estimates produced from each tree are averaged for prediction problems, and treated as votes for classification problems.

Random forests…

- Different types of forests use different approaches to random vectors Θ_k.

Random forests...

- Different types of forests use different approaches to random vectors Θ_k.
- How to compare them?

Generalization error for forests

For numerical predictors:

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$$PE^*_{\text{forest}} \leq \bar{\rho} PE^*_{\text{tree}}$$

where $\bar{\rho}$ is a weighted correlation between $Y - h(\mathsf{X},\Theta)$, $Y - h(\mathsf{X},\Theta')$ where Θ, Θ' are independent.

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where $\bar{\rho}$ is a weighted correlation between $Y - h(\mathsf{X},\Theta)$, $Y - h(\mathsf{X},\Theta')$ where Θ, Θ' are independent.
Implication: for good generalization error of a forest, we need low correlation error across Θ, and low error trees.

Some approaches to forests

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- Which approach works best depends on number of covariates, how correlated covariates are, how predictive covariates are.

- More (combinations of) features not always better.

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- Have good internal metrics of error, strength, correlation, variable importance
- Possible to parallelize?

Internal metrics

“Out-of-bag” estimation

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$$\hat{f}_i^{oob} = \frac{1}{|C^{-i}|} \sum_{b \in C^{-i}} \hat{f}^b(x_i)$$

C^{-i} is the set of bootstrap samples that do not contain i, $|C^{-i}|$ is the size of this set

Honesty.

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- As we think about causal quantities, we'll move the target.

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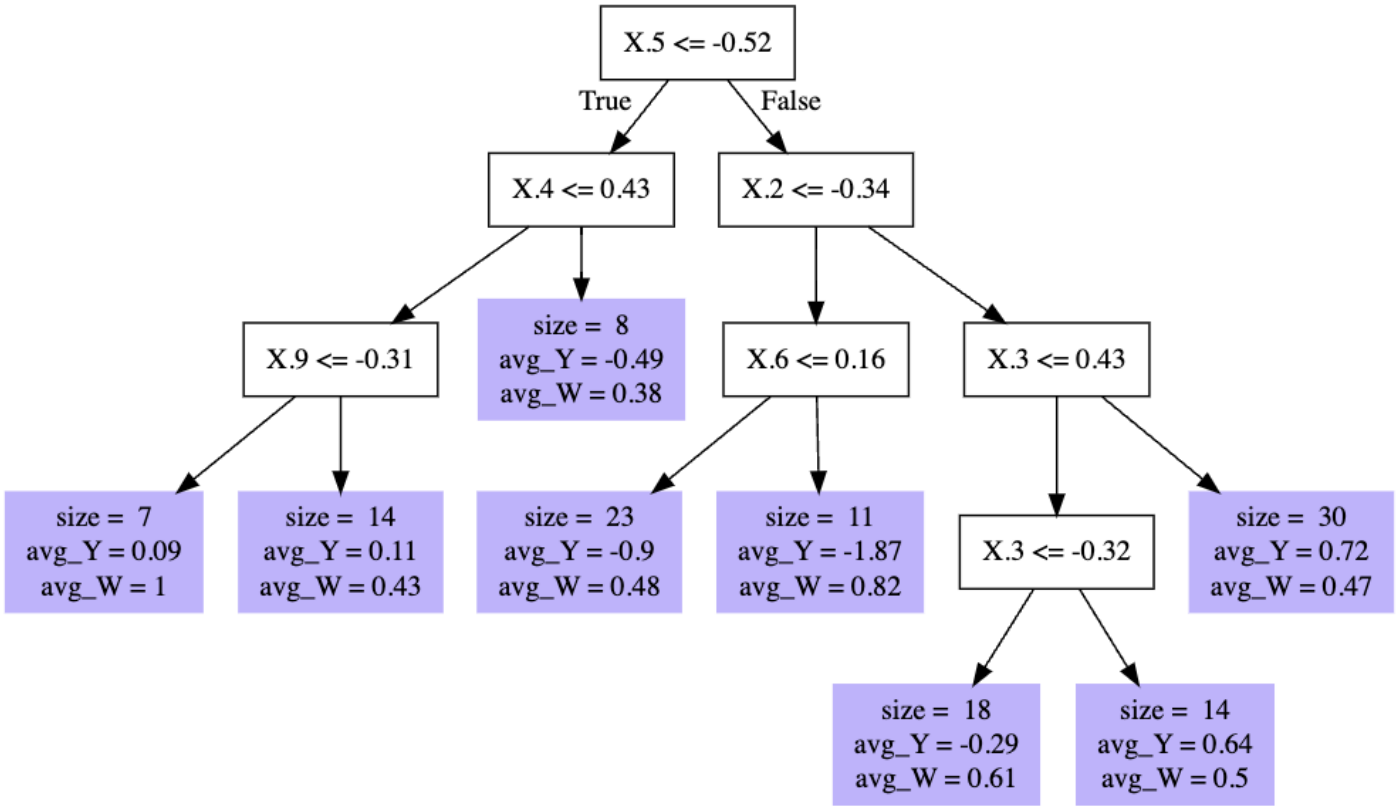
- This procedure reduces bias relative to those proposed by Breiman (2001).

A tree


X.5 <= -0.52
True
False
X.4 <= 0.43
X.2 <= -0.34
X.9 <= -0.31
size = 8
avg_Y = -0.49
avg_W = 0.38
X.6 <= 0.16
X.3 <= 0.43
size = 7
avg_Y = 0.09
avg_W = 1
size = 14
avg_Y = 0.11
avg_W = 0.43
size = 23
avg_Y = -0.9
avg_W = 0.48
size = 11
avg_Y = -1.87
avg_W = 0.82
X.3 <= -0.32
size = 30
avg_Y = 0.72
avg_W = 0.47
size = 18
avg_Y = -0.29
avg_W = 0.61
size = 14
avg_Y = 0.64
avg_W = 0.5

References I

Breiman, L. (2001). Random forests. *Machine learning*, 45:5–32.

Freund, Y., Schapire, R. E., et al. (1996). Experiments with a new boosting algorithm. In *icml*, volume 96, pages 148–156. Citeseer.

Hastie, T., Tibshirani, R., and Friedman, J. H. (2009). *The elements of statistical learning: data mining, inference, and prediction*, volume 2. Springer.